Meowbass

Short manual . free and open source (AGPL-3.0-or-later) . github.com/SpeakingColors

Meowbass is a psychoacoustic bass enhancer. It splits the bass with a phase-aligned crossover, makes harmonics from it, evens out their level and adds them back, so small speakers and headphones suggest deep bass that is not really there. It also lets you pull back the original sub to free headroom.

Quick start

- Set Bass range near the lowest note you care about (kick or bass fundamental, often 50 to 100 Hz).
- Raise Harmonics until the bass feels bigger on small speakers.
- Lower Original bass to free headroom once the harmonics carry the bass.
- Use Even/Odd to choose warm and round (even) or hard and edgy (odd).

How the window works

- Each knob sits directly under the display it changes; the displays react to the knobs and the knobs react to the displays.
- Meters show the real signal in dB with a dB scale: input and output on the far right.
- Hover a control for about one second to see an info bubble. Turn bubbles off with the round i button in the header.
- The Limiter switch in the header adds a safety limiter at the output (useful while experimenting).
- Presets are in the header; the dice picks random settings (some controls are excluded on purpose).

Controls

Frequency

Harmonic filter	Choice: Smooth, Wide, Tight (default Smooth)	Harmonic filter: Smooth is balanced, Wide lets more upper harmonics through (brighter), Tight keeps only the lowest ones (cleaner).
Bass range	30 to 200 Hz default 80 Hz	Bass range: the frequency below which the bass is picked up to make harmonics. Set it near the lowest note you want to feel (kick or bass fundamental, often 50 to 100 Hz).
Original bass	-24 to 6 dB default 0 dB	Original bass: level of the untouched low end. Pull it down to free headroom and let the harmonics carry the bass.
Harmonics	-24 to 12 dB default -6 dB	Harmonics: level of the added overtones that make small speakers suggest deep bass. Start low and raise until the bass feels bigger.

Character

Even/Odd	0 to 1 default 0.5	Even/Odd: character of the harmonics. Even = warm and round (rectifier style), Odd = harder and more edgy (clipper style).

Dynamics

Dyn ratio	1 to 10:1 default 3:1	Dynamics ratio: evens out quiet and loud bass notes so the harmonics stay steady. 1:1 follows the level exactly, higher = more even.
Dyn response	1 to 200 ms default 40 ms	Dynamics response: how fast the evening-out reacts. Short = tight and punchy, long = smooth.

Level

Input	-12 to 12 dB default 0 dB	Input: level into the effect. Raise it for more harmonics from a quiet source.
Output	-12 to 12 dB default 0 dB	Output: final level. Match it with the bypassed level when comparing.

Recommendations

- Check on small speakers or headphones: that is where the effect matters.
- Harmonic filter: Smooth for most material, Wide for brighter results, Tight for the cleanest low harmonics.
- Short Dyn response keeps kicks punchy; longer values keep sustained bass smooth.
- Do not lower Original bass so far that the club system loses real sub energy.

Presets

Init, Small Speakers, Kick Weight, Sub Clarity, Reese Body, Gentle Warmth, Hardstyle Rumble

Install and use

- Windows: run the installer (.exe). macOS: open the installer (.pkg or .dmg). Linux: install the Flatpak with `flatpak install --user` and the .flatpak file, or copy the .vst3 folder to ~/.vst3.
- Everything works offline. Rescan plugins in your DAW, then look for the plugin under SpeakingColors.
- Report problems or ideas on the GitHub page of the plugin. Free to use, copy and change under the AGPL-3.0-or-later licence.